

EDUCATION

- **Purdue University** West Lafayette, IN
 • **Ph.D. Computer Science (CIGP)**; Department of Computer Science, Advisor: Pan Li 2019 – 2025
- **Peking University** Beijing, China
 • **MPhil Data Science**; Academy for Advanced Interdisciplinary Studies 2016 – 2019
- **Shandong University** Ji'nan, China
 • **BEng Computer Science and Technology**; Taishan College* (Honors School) 2012 – 2016
 • **BEcon Finance**; School of Economics 2014 – 2016
 * Top-notch Undergraduate Training Program by Ministry of Education, China

RESEARCH INTERESTS

- Learning and Computing on Complex Structured Data
 Topics in Efficient and Scalable Graph Learning Algorithms; Large Language Models (LLMs) / Foundation Models for Multimodal Data; Trustworthy ML (Data Privacy); Spatiotemporal Data Analysis.

SELECTED PUBLICATIONS (* EQUAL CONTRIBUTIONS) [FULL LIST @ GOOGLE SCHOLAR]

- Yinan Huang*, **Haoteng Yin***, Eli Chien, Rongzhe Wei, Pan Li. Differentially Private Relational Learning with Entity-level Privacy Guarantees. *Advances in Neural Information Processing Systems 38 (NeurIPS'25)*
- Rongzhe Wei*, Peizhi Niu*, Hans Hao-Hsun Hsu, Ruihan Wu, **Haoteng Yin**, Yifan Li, Eli Chien, Kamalika Chaudhuri, Olga Milenkovic, and Pan Li. Do LLMs Really Forget? Evaluating Unlearning with Knowledge Correlation and Confidence Awareness. *Advances in Neural Information Processing Systems 38 (NeurIPS'25)*
- **Haoteng Yin**, Rongzhe Wei, Eli Chien, Pan Li. Privately Learning from Graphs with Applications in Fine-tuning Large Pretrained Models. *Conference on Language Modeling (COLM'25)*
- **Haoteng Yin**, Jinha Kim, Prashant Mathur, Krishanu Sarker, Vidit Bansal. How to Talk to Language Models: Serialization Strategies for Structured Entity Matching. *Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL'25, Findings)*
- Rongzhe Wei, Eleonora Kreačić, Haoyu Wang, **Haoteng Yin**, Eli Chien, Vamsi K Potluru, Pan Li. On the Inherent Privacy Properties of Discrete Denoising Diffusion Models. *Transactions on Machine Learning Research (TMLR'24)*
- Tianyi Zhang*, **Haoteng Yin***, Rongzhe Wei, Pan Li, Anshumali Shrivastava. Learning Scalable Structural Link Representations with Bloom Signatures. *Proceedings of the ACM Web Conference. (WWW'24)*
- Wei Xiang*, **Haoteng Yin***, He Wang, Xiaogang Jin. SocialCVAE: Predicting Pedestrian Trajectory via Interaction Conditioned Latents. *The 38th Annual AAAI Conference on Artificial Intelligence (AAAI'24)*.
- **Haoteng Yin**, Muhan Zhang, Jianguo Wang, Pan Li. SUREL+: Moving from Walks to Sets for Scalable Subgraph-based Graph Representation Learning. *Proceedings of the VLDB Endowment 16(11): 2939-2948. (VLDB'23)*
- Rongzhe Wei, **Haoteng Yin**, Junteng Jia, Austin R. Benson, Pan Li. Understanding Non-linearity in Graph Neural Networks from the Bayesian-Inference Perspective. *Advances in Neural Information Processing Systems 35 (NeurIPS'22)*
- **Haoteng Yin**, Muhan Zhang, Yanbang Wang, Jianguo Wang, Pan Li. Algorithm and System Co-design for Efficient Subgraph-based Graph Representation Learning. *Proceedings of the VLDB Endowment 15(11): 2788-2796. (VLDB'22)*
- Haorui Wang, **Haoteng Yin**, Muhan Zhang, Pan Li. Equivariant and Stable Positional Encoding for More Powerful Graph Neural Networks. *International Conference on Learning Representations (ICLR'22)*
- Bing Yu*, **Haoteng Yin***, Zhanxing Zhu. Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. *The 27th International Joint Conference on Artificial Intelligence (IJCAI'18)*.
- **YIN Haoteng** and LIU Yang. Semantic analysis of spatial temporal trajectory in LBSNs. (in Chinese) *SCIENTIA SINICA Informationis 47(8), 1051-1065(2017)*. (**SMP'16 Best Paper Award**)

INVITED TALKS & PRESENTATIONS

- (2025) Towards Scalable and Privacy-Preserving Graph Learning via System-aware Algorithm Design. *Artificial Intelligence Initiative (Aii) @ University of Central Florida*, Orlando, FL [Link]
- (2025) Algorithm-System Co-Design for Scalable and Privacy-Preserving Graph Learning. *PurPL Seminar*, Hosted by Prof. Ben Delaware, W. Lafayette, IN [Link]
- (2025) Privately Learning from Graphs with Applications in Fine-tuning Large Language Models. *IBM Global Digital Health*, Hosted by Dr. Bo Wen, Online
- (2024) Privately Learning from Graphs with Applications in Fine-tuning Large Pretrained Models. *The 3rd RCAC Annual Cyberinfrastructure Symposium*, W. Lafayette, IN [**Best Poster Award, News**]
- (2024) Algorithm and System Co-design for Scalable and Efficient Subgraph-based Graph Representation Learning. *Learning on Graphs & Geometry Reading Group*, Online [Link][Video]
- (2023) SUREL+: Moving from Walks to Sets for Scalable Subgraph-based Graph Representation Learning. *The 2nd RCAC Annual Cyberinfrastructure Symposium, CI + AI 2023*, W. Lafayette, IN
- (2023) Towards Scalable and Efficient Subgraph-based Graph Representation Learning. *Amazon A9 Machine Learning Reading Group*, Palo Alto, CA
- (2019) Spatio-Temporal Graph Prediction through Deep Graph Neural Networks. *The 3rd Peking University Graduate Student Forum on Advanced Interdisciplinary Studies*, Beijing, China [Video][**Best Oral Presentation Award**]

ACADEMIC EXPERIENCE

- **Purdue University** West Lafayette, IN
Graduate Assistantship, under the supervision of Prof. Pan Li. Aug. 2019 - May. 2025
 - **Differentially Private Relational Learning**
Design privacy-preserving relational learning frameworks for network-structured data, with application in fine-tuning large language models (e.g., Llama2-7B) on sensitive graphs to be relation-aware with entity- and relation-level privacy guarantees.
 - **Subgraph-based Graph Representation Learning (SGRL)**
Design novel frameworks SUREL/SUREL+ for subgraph-based GRL through system-aware algorithm design. A decoupled paradigm is proposed to sample and encode random walks and node sets for representation learning over subgraphs, which enables efficient training and inference on large-scale graphs for link, relation, and higher-order prediction tasks. SUREL/SUREL+ ranks in the Top 5 on Open Graph Benchmark.
- **LIX, École Polytechnique & Inria** Palaiseau, France
Research Intern, under the supervision of Prof. Catuscia Palamidessi. Jun. 2018 - Sep. 2018
 - **Machine Learning Attacks to Location Privacy**
Develop an adversary framework that uses sequence models to capture users' mobility patterns from obfuscated trajectory data, which are gathered from location-based services and protected by a privacy mechanism, and then perform re-identification attacks to measure the privacy vulnerability.
- **Center for Data Science, Peking University** Beijing, China
Graduate Research Assistant, under the supervision of Prof. Zhanxing Zhu Jan. 2017 - Aug. 2019
 - **Traffic Prediction with Deep Spatial Temporal Graph Neural Networks**
Design a deep learning framework, STGCN, to predict traffic time series, which coherently models the topology of road networks and captures periodic traffic patterns through spatiotemporal convolutional blocks. It has become a prominent model for network time series and is widely used in transportation research (5.8K+ citations).

INDUSTRIAL EXPERIENCE

- **Amazon AGI** Bellevue, WA
Applied Scientist Intern, Mentor: Dr. Jinha Kim May. 2024 - Aug. 2024
 - **Cross-Domain Entity Resolution with Graph-Enriched Language Models**
- **Amazon Search** Palo Alto, CA
Applied Scientist Intern, Mentors: Shweta Garg and Dr. Changhe Yuan Jun. 2023 - Sep. 2023
 - **Graph Foundation Models for Large-scale Text-attributed E-commerce Network**

PROFESSIONAL SERVICES

- Conference Reviewer
 - NeurIPS'22-25, ICML'22-25, ICLR'24-26, WWW'22,24, WSDM'23, LoG'22-23, CISS'23
- Conference Program Committee Member
 - **IEEE** SPAWC'21, BigData'25
- Journal Reviewer
 - **Nature** Machine Intelligence;
 - **ACM** Transactions on Spatial Algorithms and Systems;
 - **IEEE** Transactions on Big Data, Cybernetics, Emerging Topics in Computing, Knowledge and Data Engineering, Information Forensics & Security, Intelligent Transportation Systems, Internet of Things Journal, Neural Networks and Learning Systems, Signal Processing
- Graduate Teaching Assistant, Purdue University
 - CS242 Introduction to Data Science - 19'Fall
 - CS243 Artificial Intelligence Basics - 24'Fall
 - CS38003 Python Programming - 20'Spring, 23'Spring
 - CS490-LDA Large Scale Data Analytics - 20'Fall
 - CS590-DE I&II Data Engineering - 20'Spring, 20'Summer, 21'Spring (*Lead)
 - CS590-NCDS Numerical Computing for Data Science - 21'Spring
 - CS593-CLG Computation & Learn on Graphs - 22'Fall (Guest Lecture: Build a Graph Learning Pipeline with PyG on Colab)
- Poster Session Judge - Purdue Spring Undergraduate Research Conference *Apr. 2024, 2025*
- Graduate Mentor - Purdue CS Graduate Student Board Mentor-Mentee Program *Aug. 2021, 2024*
- Student Volunteer - The International Machine Learning Society *Jul. 2018*
- Student Membership - AAI, ACM, SIAM

SELECTED HONORS & AWARDS

- AMiner AI 2000 Most Influential Scholar Award (Honorable Mention #46) in AAI/IJCAI *Jul. 2025*
- The 3rd RCAC Cyberinfrastructure Symposium Best Poster Award (\$1,000) *Oct. 2024*
- Purdue Teaching Academy Graduate Teaching Award (2%) *Apr. 2024*
- AMiner AI 2000 Most Influential Scholar Award (Honorable Mention #60) in AAI/IJCAI *Mar. 2024*
- The Top Reviewer Award (8%) for NeurIPS'22 *Oct. 2022*
- Graduate School Summer Research Grants, Purdue University *Mar. 2021*
- Academic Excellence Scholarship, Peking University (~8%) *Oct. 2016, 2017, 2018*
- The 2nd DiDi-IEEE Elite Forum Extraordinary Potential Award (2/39) *May. 2018*
- École Polytechnique International Student (Intern) Fellowship *Mar. 2018*
- The 5th National Conference of Social Media Processing the Best Paper Award (1/25) *Oct. 2016*
- Taishan College International Study Scholarship (CNY¥33,000) *Jun. 2015*